

Learning outcomes	Additional guidance
<i>Learners should be able to demonstrate and apply their knowledge and understanding of:</i> (a) systematic errors (including zero errors) and random errors in measurements (b) precision and accuracy (c) absolute and percentage uncertainties when data are combined by addition, subtraction, multiplication, division and raising to powers (d) graphical treatment of errors and uncertainties; line of best fit; worst line; absolute and percentage uncertainties; percentage difference.	As discussed in <i>The Language of Measurement</i> (ASE 2010). A rigorous statistical treatment is not expected. <i>M1.5</i> An elementary knowledge of error bars is expected at A level. <i>M1.5</i>